

DATABRICKS LAKEHOUSE VS HADOOP MAPREDUCE: A COMPARATIVE ANALYSIS OF IN-MEMORY PROCESSING AND DATA RELIABILITY FOR MODERN BIG DATA ANALYTICS

Nurul Adriana Binti Kamal Jefri

Faculty of Computing, Universiti Teknologi Malaysia, Johor Bahru
adriana.kjefri@gmail.com

ABSTRACT

Big data systems, like Hadoop MapReduce, face performance limitations as a result of disk-based processing, resulting in high I/O overheads and low data reliability. These challenges are increasingly significant as organisations seek to leverage scalable, interactive analytics and reliable data to gain insights from large data sets (Gandomi & Haider, 2014). In this paper, we compare the Databricks Lakehouse model with traditional Hadoop MapReduce platform in terms of in-memory data integrity and processing speed. The results reveal that the Lakehouse model overcomes the traditional constraints by utilising the Apache Spark engine for in-memory processing and Delta Lake storage layer for ACID transactions (Armbrust et al., 2020; Zaharia et al., 2016). By delivering up to 100x faster performance on iterative jobs (Shi et al., 2015), the Lakehouse approach comes with its own limitations, such as increased memory consumption and greater system complexity (Kumar & Li, 2022). The Databricks Lakehouse is an important advancement in the big data ecosystem for companies seeking a combination of compute performance and data reliability.

Keywords: Databricks Lakehouse, Hadoop MapReduce, in-memory processing, big data analytics, Delta Lake

1.0 INTRODUCTION

With the rapid increase in data within enterprises, the need for scalable analytics to extract insights from data in near real-time has been magnified (Gandomi & Haider, 2014). This requires platforms capable of efficiently handling large volumes of data while offering high throughput and low latency to support a variety of analytical techniques.

Yet classical batch processing systems struggle to address these demands. Existing frameworks like Hadoop MapReduce are disk-intensive, resulting in significant I/O bottlenecks during iterative processing or complex queries, thus reducing performance (Dean & Ghemawat, 2009). Additionally, even though the Hadoop Distributed File System (HDFS) is a critical component of data storage, it remains a reliability and maintenance challenge for today's fast-moving workloads (Shvachko et al., 2010).

As such, the need to adopt a platform that combines speed, reliability and scalability drives interest among contemporary enterprises. It is essential to transition from fragmented data environments (with separate data lakes and data warehouses) to a system that combines the performance of in-memory processing with robust reliability (Zaharia et al., 2021b).

In this paper, we benchmark and compare the Databricks Lakehouse system against the traditional Hadoop MapReduce system. The aim is to compare these systems' in-memory processing capabilities and reliability. The content of this paper is divided into sections: Section 2 gives a technical review of pertinent technologies; Section 3 compares the architecture and benchmarking performance; and Section 4 offers the conclusion.

2.0 LITERATURE REVIEW

High-performance data processing had its origins in distributed batch processing. Dean and Ghemawat (2009) initiated the programming model of MapReduce to enable massive parallel processing but with the constraint of intermediate disk storage. This design constraint was further exacerbated by the Hadoop Distributed File

System (HDFS), which, although providing the scalability needed in big data, was primarily oriented to high bandwidth reads rather than low latency random access needed for interactive and iterative processing (Shvachko et al., 2010).

Zaharia et al. (2016) built on these limitations to introduce the next generation of big data engine Apache Spark with minimised disk I/O via in-memory processing. Spark's in-memory caching addressed the limitation of the MapReduce framework to effectively handle the iterative machine learning and data mining workloads, and paved the way for high-performance processing. However, while processing performance improved, there remained a gap between the cost-effective data lakes and the trusted data warehouses.

To overcome this, Zaharia et al. (2021b) proposed the Lakehouse that merges the open storage of lakes with the transactional guarantees of warehouses. This concept was realised in Delta Lake that offers ACID transactions and metadata management on open cloud object storage (Armbrust et al., 2020). This evolution to holistic and trusted systems forms the context of our architectural and performance analysis in the next section.

3.0 ARCHITECTURAL & PERFORMANCE ANALYSIS

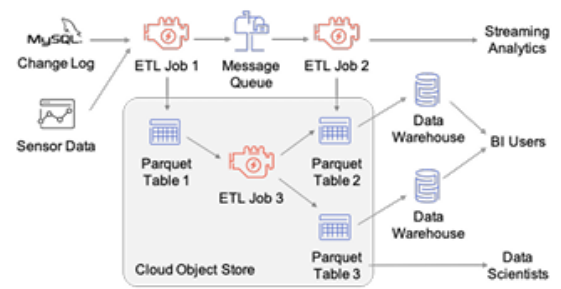
This chapter assesses the evolution of distributed systems to integrated platforms through architectural efficiency, processing speed and practical applications.

3.1 ARCHITECTURE OVERVIEW

The Hadoop ecosystem implemented a strictly integrated master-slave architecture for disk-bound batch computation. Here the NameNode stores metadata and DataNodes store data on disks. Processing using MapReduce includes a disk-based shuffle phase, with intermediate data written to disk between processing stages, resulting in I/O bottlenecks (Shvachko et al., 2010).

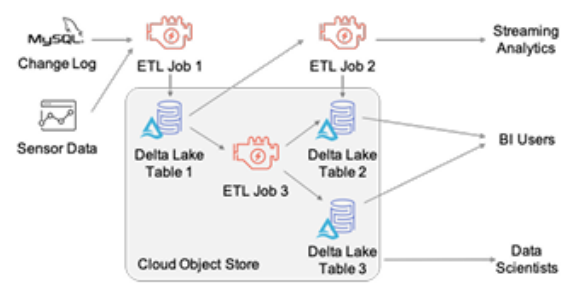
The Databricks Lakehouse is built on a decoupled model that decouples storage from computing resources. In this approach, depicted in Figure 1 and Figure 2, data is stored in open-source formats on cloud object storage and computation is done in memory by the Apache Spark engine (Kumar & Li, 2022). The Delta Lake

storage layer is a critical component, serving as a buffer for reliability between the data lake and analytics engine (Armbrust et al., 2020).



(a) Pipeline using separate storage systems.

Figure 1 Separate storage pipeline (Armbrust et al., 2020).



(b) Using Delta Lake for both stream and table storage.

Figure 2 Delta Lake unified pipeline (Armbrust et al., 2020).

3.2 COMPARATIVE ANALYSIS

The advantages of modern unified data platforms over traditional data platforms are illustrated with a comparative analysis of major functional areas.

The differences are outlined in Table 1 and supported by empirical studies;

Table 1 Comparison between Hadoop MapReduce and the Databricks lakehouse platform (Shvachko et al., 2010)

Criteria	Hadoop MapReduce	Databricks Lakehouse
Processing Model	Disk-based, batch only	In-memory via Spark engine
Speed	High I/O latency	Up to 100x faster for iterative jobs
Data Reliability	No ACID support	ACID transactions via Delta Lake
Storage	HDFS tightly	Decoupled

Model	coupled	object storage
Scalability	Vertical scaling limitations	Elastic cloud-native scaling
Use Case Fit	Simple large-scale batch ETL	Batch, streaming, and ML workloads

Benchmark tests by Shi et al. (2015) confirm that Spark systems like Databricks are significantly faster (orders of magnitude) than disk-based MapReduce systems for complex iterative applications.

3.3 REAL-WORLD APPLICATION

The performance benefits of the Lakehouse model are seen in real-time analytical scenarios, such as stock market analysis. Yasmin and Kamalakkannan (2022) show that by adopting Databricks Delta Lake for stock market data, queries become more efficient than those on conventional data lakes. The enhanced file management and indexing features of Delta Lake help organizations conduct low latency time-series analysis on large volumes of financial data of diverse types while fulfilling stringent reliability requirements.

4.0 CONCLUSION

This paper has assessed the architectural and performance evolution from batch to unified data platforms. The paper shows the Databricks Lakehouse platform successfully overcomes the inherent disk I/O and reliability constraints of Hadoop MapReduce by combining the high-performance in-memory capabilities of the Spark engine with ACID-compliant storage of Delta Lake (Armbrust et al., 2020; Zaharia et al., 2021b). This transition has practical implications in the sense that companies with iterative, real-time and/or reliable processing requirements can benefit from substantial performance improvements and simplification of operations when transitioning from traditional Hadoop ecosystems to the Lakehouse model.

But the use of such advanced systems comes with limitations. Lakehouse platforms tend to be more complex and expensive in terms of memory costs than pure MapReduce platforms, and require expert technical skills for efficient administration (Kumar & Li, 2022). Thus, future work should explore native integration of AI/ML with Lakehouse platforms and petabyte-scale cost-performance optimisation, as these will impact the future adoption of unified models for enterprise analytics.

REFERENCES

- Armbrust, M., Das, T., Sun, L., Yavuz, B., Zhu, S., Murthy, M., Torres, J., Van Hovell, H., Ionescu, A., Łuszczak, A., Świtakowski, M., Szafranski, M., Li, X., Ueshin, T., Mokhtar, M., Boncz, P., Ghodsi, A., Paranjpye, S., Senster, P., . . . Zaharia, M. (2020). Delta lake. *Proceedings of the VLDB Endowment*, 13(12), 3411–3424. <https://doi.org/10.14778/3415478.3415560>
- Dean, J., & Ghemawat, S. (2009). MapReduce. *Communications of the ACM*, 53(1), 72–77. <https://doi.org/10.1145/1629175.1629198>
- Gandomi, A., & Haider, M. (2014). Beyond the hype: Big data concepts, methods, and analytics. *International Journal of Information Management*, 35(2), 137–144. <https://doi.org/10.1016/j.ijinfomgt.2014.10.007>
- Kumar, D., & Li, S. (2022). Separating Storage and Compute with the Databricks Lakehouse Platform. *2022 IEEE 9th International Conference on Data Science and Advanced Analytics (DSAA)*, 1–2. <https://doi.org/10.1109/dsaa54385.2022.10032386>
- Shi, J., Qiu, Y., Minhas, U. F., Jiao, L., Wang, C., Reinwald, B., & Özcan, F. (2015). Clash of the titans. *Proceedings of the VLDB Endowment*, 8(13), 2110–2121. <https://doi.org/10.14778/2831360.2831365>
- Shvachko, K., Kuang, H., Radia, S., & Chansler, R. (2010). The Hadoop distributed file system. *2010 IEEE 26th Symposium on Mass Storage Systems and Technologies*, 1–10. <https://doi.org/10.1109/msst.2010.5496972>
- Yasmin, A., & Kamalakkannan, S. (2022). Analytical performance in Data Lake storage of big data analytics by DataBricks Delta Lake for stock market analysis. In *Sustainable Expert Systems* (Vol. 133, pp. 227–241). Springer Nature. https://doi.org/10.1007/978-981-19-3035-5_17

- Zaharia, M., Ghodsi, A., Xin, R., & Armbrust, M. (2021b). Lakehouse: A New Generation of Open Platforms that Unify Data Warehousing and Advanced Analytics. *Conference on Innovative Data Systems Research*. http://cidrdb.org/cidr2021/papers/cidr2021_paper17.pdf
- Zaharia, M., Xin, R. S., Wendell, P., Das, T., Armbrust, M., Dave, A., Meng, X., Rosen, J., Venkataraman, S., Franklin, M. J., Ghodsi, A., Gonzalez, J., Shenker, S., & Stoica, I. (2016). Apache Spark. *Communications of the ACM*, 59(11), 56–65. <https://doi.org/10.1145/2934664>

LOGBOOK

Date	Search Keywords	AI Prompts used	Task Description
Apr 23, 2026	Subject assignment briefing, Big Data Analytics topics	"Can you guide me on how I should start?"	Reviewed the assignment briefing and selected Topic 12 (Big Data Analytics). Choose Databricks as the focus technology for comparison with Hadoop MapReduce.
Apr 24, 2026	Databricks vs Snowflake Scopus, Armbrust Lakehouse Scopus index	"Let's do databricks, is it hard? can i compare with jupyter notebook?", "look into these references that I found and analyze if it suits"	Searched for and verified Scopus-indexed references. Replaced non-indexed sources with peer-reviewed journals to meet the minimum requirement of 7 references.
Apr 25, 2026	Databricks Lakehouse performance Scopus, Decoupled storage vs compute comparison	"Let's start with an introduction first, what should I write?", "review if the literature review I wrote are according to the specific requirement"	Drafted the Introduction and Literature Review. Focused on the narrative progression from MapReduce to Spark, and finally to the Lakehouse architecture.
Apr 26, 2026	Analytical performance Databricks Delta Lake stock market	"Did this section that I wrote follow the specific requirements for this section?", "can i use this photo as figure 1?"	Wrote the main body subtopics, including the Architecture Overview and Comparative Analysis table. Adapted an architecture diagram and cited it correctly as Figure 1.
Apr 27, 2026	N/A (Final Editing)	"Does the conclusion cover all the sections before it", "review this fixed conclusion", "does the abstract sound professional?"	Finalized the Conclusion and Abstract. Performed a final citation audit to ensure all technical claims were evidence-based and formatted the reference list in APA 7th Edition style.